BS-M301 — PROBABILITY AND STATISTICS

MODULE 6: SMALL SAMPLES

B.Tech ECE — 3rd Semester | MAKAUT Syllabus: 4L

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1. Introduction to Small Sample Theory

Definition: Small Samples vs. Large Samples

In statistical hypothesis testing, a sample is designated as a **small sample** if its sample size $n < 30$.

- **Large Samples** ($n \geq 30$)**:** By the Central Limit Theorem (CLT), the sampling distribution of sample means approaches normality regardless of the parent

distribution. The sample standard deviation s closely approximates the population standard deviation σ, justifying the use of the standard normal Z-test.

- **Small Samples ($n < 30$):** The CLT cannot be invoked. The parent population **must be assumed normally distributed**. Furthermore, replacing σ with the sample variance $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$ introduces substantial additional sampling fluctuations, yielding distributions with heavier tails than $N(0,1)$.

Hence, small sample inference relies on the **exact sampling distributions**:

1. **Student's t-distribution** (W. S. Gosset, 1908): For inference on population means and correlation coefficients when σ is unknown.
2. **Snedecor's F-distribution** (G. W. Snedecor & R. A. Fisher): For testing the ratio of two independent population variances.
3. **Chi-square (χ^2) distribution** (Karl Pearson, 1900): For tests of goodness of fit and independence of attributes in contingency tables.

Degrees of Freedom (ν or d.f.)

The **degrees of freedom** (ν) of a test statistic represents the number of unconstrained, freely variable observations available in the calculation:

$$\nu = n - k \tag{1}$$

where n is the total number of sample observations and k is the number of linear constraints or sample-derived parameters estimated to compute the statistic (e.g., estimating μ by $\bar{x}$ consumes 1 d.f., so $\nu = n - 1$).

2. Student's t-Distribution

2.1 Definition and Derivation

Mathematical Definition of Student's t-Variable

Let $Z \sim N(0,1)$ be a standard normal random variable, and let $Y \sim \chi^2(\nu)$ be an independent Chi-square random variable with ν degrees of freedom. The ratio:

$$T = \frac{Z}{\sqrt{\frac{Y}{\nu}}} \tag{2}$$

follows **Student's t-distribution** with ν degrees of freedom, written $T \sim t(\nu)$.

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2.2 Probability Density Function and Properties

The probability density function (PDF) of $T \sim t(\nu)$ is:

$$f(t) = \frac{1}{\sqrt{\nu}\,\mathrm{B}\left(\frac{1}{2}, \frac{\nu}{2}\right)} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad -\infty < t < \infty \tag{3}$$

where $\mathrm{B}\left(\frac{1}{2}, \frac{\nu}{2}\right) = \frac{\sqrt{\pi}\,\Gamma(\nu/2)}{\Gamma((\nu+1)/2)}$ is the Beta function.

Key Properties of Student's t-Distribution

1. **Symmetry:** The curve is bell-shaped and symmetric about $t = 0$. All odd central moments are zero:

$$\mu_{2k+1} = E[T^{2k+1}] = 0 \quad (\text{for } \nu > 2k+1) \tag{4}$$

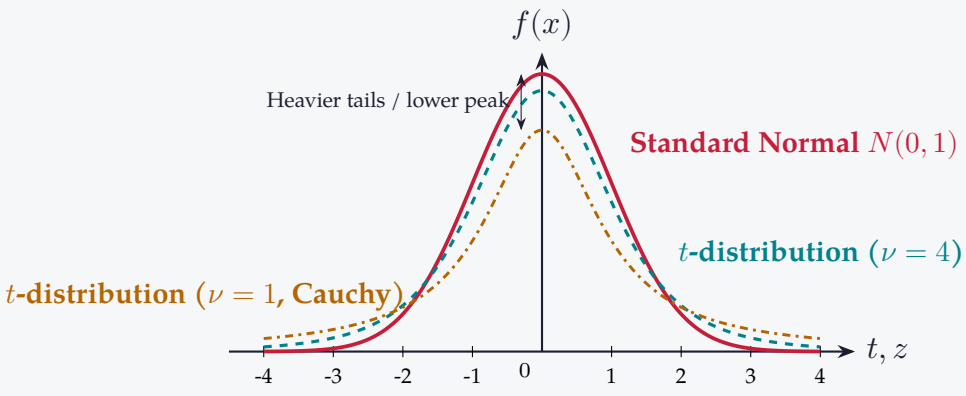
2. **Mean:** $\mu_1' = E[T] = 0$ (valid for $\nu > 1$).
3. **Variance:** $\mathrm{Var}(T) = \mu_2 = \dfrac{\nu}{\nu - 2}$ (valid for $\nu > 2$). Note that $\mathrm{Var}(T) > 1$.
4. **Kurtosis (Heavy Tails):** The distribution is leptokurtic ($\beta_2 > 3$). For $\nu > 4$:

$$\beta_2 = \frac{\mu_4}{\mu_2^2} = 3 + \frac{6}{\nu - 4} > 3 \tag{5}$$

5. **Limiting Form (Asymptotic Normality):** As $\nu \to \infty$, the t-distribution converges to the standard normal distribution:

$$\lim_{\nu\to\infty} \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}} = e^{-t^2/2} \implies t(\nu) \xrightarrow{d} N(0,1) \tag{6}$$

TikZ Illustration: Comparison of Student's t vs. Standard Normal $N(0,1)$



3. Student's t-Tests of Significance

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3.1 Test for Single Mean

Test 1: Single Mean t-Test Formulation

Let $x_1, x_2, \ldots, x_n$ be a random sample of size $n < 30$ drawn from a normal population $N(\mu, \sigma^2)$ where σ is unknown. We test:

$$H_0 : \mu = \mu_0 \quad \text{vs.} \quad H_1 : \mu \neq \mu_0 \quad (\text{or one-sided } \mu > \mu_0, \mu < \mu_0) \tag{7}$$

The test statistic is:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \tag{8}$$

where $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ and the **unbiased sample variance** is:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} x_i^2 - \frac{(\sum x_i)^2}{n}\right) \tag{9}$$

Under H_0, $t \sim t(\nu)$ with degrees of freedom $\nu = n - 1$.
Decision Rule (Two-tailed at significance level α):

- Reject H_0 if $|t_{\text{cal}}| > t_{\alpha/2, n-1}$.
- Fail to reject H_0 if $|t_{\text{cal}}| \leq t_{\alpha/2, n-1}$.

$100(1-\alpha)\%$ **Confidence Interval for μ:**

$$\bar{x} - t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} \leq \mu \leq \bar{x} + t_{\alpha/2,n-1}\frac{s}{\sqrt{n}} \tag{10}$$

Worked Example: Single Mean t-Test

A manufacturer claims that the mean lifetime of an electronic component is 1000 hours. A random sample of 9 components yields a sample mean of 980 hours with a sample standard deviation $s = 30$ hours. Test at the 5% level whether the true mean lifetime is significantly less than 1000 hours ($t_{0.05,8} = 1.860$).
Solution:

1. **Hypotheses:** $H_0 : \mu = 1000$ vs. $H_1 : \mu < 1000$ (one-tailed left-sided test).
2. **Parameters:** $n = 9$, $\bar{x} = 980$, $s = 30$, $\mu_0 = 1000$.
3. **Test Statistic:**
$$t_{\text{cal}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{980 - 1000}{30/\sqrt{9}} = \frac{-20}{10} = -2.00$$
4. **Critical Value:** For left-tailed test at $\alpha = 0.05$ with $\nu = 8$: $t_{\text{crit}} = -1.860$.
5. **Conclusion:** Since $t_{\text{cal}} = -2.00 < -1.860$, t_{cal} falls in the critical rejection region. We reject H_0 and conclude that the mean lifetime is significantly less than 1000 hours.

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3.2 Test for Difference of Two Means (Independent Samples)

Test 2: Two-Sample Independent t-Test with Pooled Variance

Let two independent samples of sizes n_1 and n_2 be drawn from normal populations $N(\mu_1, \sigma^2)$ and $N(\mu_2, \sigma^2)$ with **equal unknown variance** ($\sigma_1^2 = \sigma_2^2 = \sigma^2$). We test:

$$H_0 : \mu_1 = \mu_2 \quad \text{vs.} \quad H_1 : \mu_1 \neq \mu_2 \tag{11}$$

The **pooled unbiased variance** s_p^2 combining both samples is:

$$s_p^2 = \frac{\sum_{i=1}^{n_1}(x_{1i} - \bar{x}_1)^2 + \sum_{j=1}^{n_2}(x_{2j} - \bar{x}_2)^2}{n_1 + n_2 - 2} = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2} \tag{12}$$

The test statistic is:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{\dfrac{1}{n_1} + \dfrac{1}{n_2}}} \tag{13}$$

Under H_0, $t \sim t(\nu)$ with degrees of freedom $\nu = n_1 + n_2 - 2$.

3.3 Paired t-Test for Dependent Samples

Test 3: Paired t-Test Principle

When observations in two samples are drawn in matched pairs (e.g., measurements taken on the **same physical circuit before and after** temperature cycling), the two groups are statistically dependent. We eliminate inter-subject variability by evaluating individual differences:

$$d_i = x_i - y_i, \quad i = 1, 2, \ldots, n \tag{14}$$

Under the null hypothesis of no treatment effect ($H_0 : \mu_d = 0$):

$$t = \frac{\bar{d} - 0}{s_d/\sqrt{n}} \tag{15}$$

where:

$$\bar{d} = \frac{1}{n}\sum_{i=1}^{n} d_i, \qquad s_d^2 = \frac{1}{n-1}\sum_{i=1}^{n}(d_i - \bar{d})^2 = \frac{\sum d_i^2 - \dfrac{(\sum d_i)^2}{n}}{n-1} \tag{16}$$

The statistic follows Student's t-distribution with $\nu = n - 1$ degrees of freedom.

3.4 Test for Significance of Correlation Coefficient

Test 4: Significance of Correlation Coefficient

Let r be the Pearson sample correlation coefficient calculated from n pairs of observations drawn from a bivariate normal population. To test:

$$H_0 : \rho = 0 \quad \text{(variables are uncorrelated)} \quad \text{vs.} \quad H_1 : \rho \neq 0 \tag{17}$$

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The test statistic is:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \tag{18}$$

Under H_0, $t \sim t(\nu)$ with degrees of freedom $\nu = n - 2$. Reject H_0 if $|t_{\text{cal}}| > t_{\alpha/2,n-2}$.

4. Snedecor's F-Distribution and Test for Ratio of Variances

4.1 Definition and Properties

Definition of Snedecor's F-Distribution

Let $U \sim \chi^2(\nu_1)$ and $V \sim \chi^2(\nu_2)$ be two independent Chi-square random variables with ν_1 and ν_2 degrees of freedom, respectively. The ratio:

$$F = \frac{U/\nu_1}{V/\nu_2} \tag{19}$$

follows **Snedecor's F-distribution** with (ν_1, ν_2) degrees of freedom, denoted $F \sim F(\nu_1, \nu_2)$.

Properties of the F-Distribution

1. **Support and Shape:** $F \in (0, \infty)$. The distribution is positively skewed (right-skewed) and unimodal.
2. **Mean:** $E[F] = \dfrac{\nu_2}{\nu_2 - 2}$ (for $\nu_2 > 2$, independent of ν_1).
3. **Variance:** $\text{Var}(F) = \dfrac{2\nu_2^2(\nu_1 + \nu_2 - 2)}{\nu_1(\nu_2 - 2)^2(\nu_2 - 4)}$ (for $\nu_2 > 4$).
4. **Reciprocal Property:** If $F \sim F(\nu_1, \nu_2)$, then $\dfrac{1}{F} \sim F(\nu_2, \nu_1)$. Hence:

$$F_{1-\alpha,\nu_1,\nu_2} = \frac{1}{F_{\alpha,\nu_2,\nu_1}} \tag{20}$$

5. **Relation to t and χ^2:**

$$F_{1,\nu} = t_\nu^2, \qquad \lim_{\nu_2 \to \infty} \nu_1 F_{\nu_1,\nu_2} = \chi^2(\nu_1) \tag{21}$$

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4.2 F-Test for Equality of Two Population Variances

F-Test Procedure for $\sigma_1^2 = \sigma_2^2$

Let two independent random samples of sizes n_1 and n_2 be drawn from normal populations $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$. We test:

$$H_0 : \sigma_1^2 = \sigma_2^2 \quad \text{vs.} \quad H_1 : \sigma_1^2 \neq \sigma_2^2 \quad (\text{or } \sigma_1^2 > \sigma_2^2) \tag{22}$$

Compute the unbiased sample variances:

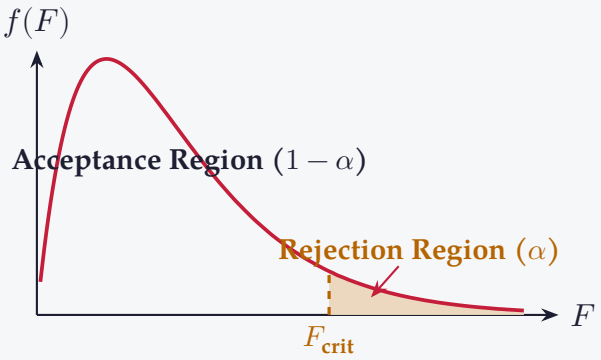
$$s_1^2 = \frac{1}{n_1 - 1} \sum_{i=1}^{n_1} (x_{1i} - \bar{x}_1)^2, \qquad s_2^2 = \frac{1}{n_2 - 1} \sum_{j=1}^{n_2} (x_{2j} - \bar{x}_2)^2 \tag{23}$$

By convention, we place the larger sample variance in the numerator:

$$F_{\text{cal}} = \frac{s_1^2}{s_2^2} \geq 1 \quad (\text{assuming } s_1^2 \geq s_2^2) \tag{24}$$

with degrees of freedom $\nu_1 = n_1 - 1$ (numerator) and $\nu_2 = n_2 - 1$ (denominator). Reject H_0 at level α if $F_{\text{cal}} > F_{\alpha/2, \nu_1, \nu_2}$.

TikZ Schematic: Right-Skewed F-Distribution with Critical Region



5. Chi-Square (χ^2) Distribution

5.1 Definition and Properties

Definition of Chi-Square Distribution

If $Z_1, Z_2, \ldots, Z_\nu$ are ν independent standard normal random variables ($Z_i \sim N(0,1)$), the sum of their squares:

$$X = \sum_{i=1}^{\nu} Z_i^2 \tag{25}$$

follows a **Chi-square distribution** with ν degrees of freedom, denoted $X \sim \chi^2(\nu)$.

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Mathematical Properties of $\chi^2(\nu)$

1. **Probability Density Function:**

$$f(x) = \frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\frac{\nu}{2}-1} e^{-x/2}, \quad x \geq 0 \tag{26}$$

2. **Mean and Variance:**

$$E[X] = \nu, \qquad \text{Var}(X) = 2\nu \tag{27}$$

3. **Moment Generating Function (MGF):**

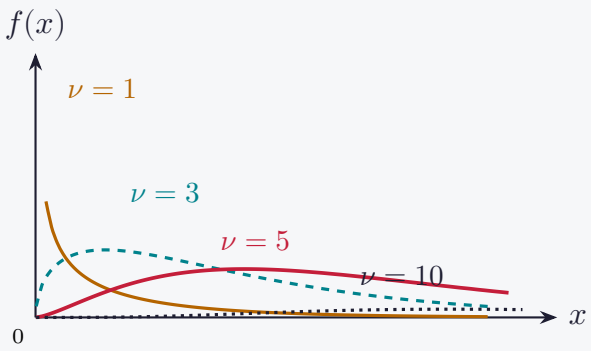
$$M_X(t) = (1-2t)^{-\nu/2}, \quad \text{for } t < \frac{1}{2} \tag{28}$$

4. **Additive (Reproductive) Property:** If $X_1 \sim \chi^2(\nu_1)$ and $X_2 \sim \chi^2(\nu_2)$ are independent:

$$X_1 + X_2 \sim \chi^2(\nu_1 + \nu_2) \tag{29}$$

5. **Limiting Form:** For large ν ($\nu > 30$), $\sqrt{2\chi^2} - \sqrt{2\nu - 1} \approx N(0,1)$.

TikZ Plot: Chi-Square Curves for Degrees of Freedom $\nu = 1, 3, 5, 10$



6. Chi-Square Test of Goodness of Fit

6.1 Principle and Pearson's Test Statistic

Pearson's Chi-Square Goodness of Fit

The goodness of fit test evaluates whether an observed frequency distribution conforms to a theoretical probability distribution (such as Binomial, Poisson, or Normal). Let:

- O_i = Observed frequency in the i-th class ($i = 1, 2, \ldots, k$).
- E_i = Expected frequency under theoretical model, with $\sum_{i=1}^{k} E_i = \sum_{i=1}^{k} O_i = N$.

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Karl Pearson proved that the test statistic:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} = \sum_{i=1}^{k} \frac{O_i^2}{E_i} - N \tag{30}$$

asymptotically follows a Chi-square distribution with ν degrees of freedom:

$$\nu = k - 1 - m \tag{31}$$

where k is the number of classes after pooling, and m is the number of independent population parameters estimated from the sample data.

Conditions for Validity of χ^2 Test

To ensure accurate approximation of discrete multinomial probabilities by continuous χ^2:

1. **Total Sample Size:** $N = \sum O_i \geq 50$.
2. **Minimum Expected Frequency:** Each $E_i \geq 5$. If any $E_i < 5$, it must be **pooled (combined)** with adjacent classes, reducing k.
3. **Independent Observations:** Sampling units must be mutually independent.
4. **Linear Constraint:** $\sum O_i = \sum E_i = N$, which consumes 1 degree of freedom.

Worked Example: Chi-Square Test of Goodness of Fit

A die is rolled 120 times, yielding the following face frequencies: 15, 23, 18, 16, 26, 22. Test at the 5% level whether the die is fair ($\chi^2_{0.05,5} = 11.07$).
Solution:

1. **Hypotheses:** H_0: The die is fair ($p_i = 1/6$) vs. H_1: The die is biased.
2. **Expected Frequencies:** $E_i = 120 \times \frac{1}{6} = 20$ for all $i = 1, \ldots, 6$. All $E_i \geq 5$.
3. **Computation Table:**

Face (i)	Observed (O_i)	Expected (E_i)	$\mathbf{O_i - E_i}$	$\mathbf{(O_i - E_i)^2/E_i}$
1	15	20	-5	$25/20 = 1.25$
2	23	20	$+3$	$9/20 = 0.45$
3	18	20	-2	$4/20 = 0.20$
4	16	20	-4	$16/20 = 0.80$
5	26	20	$+6$	$36/20 = 1.80$
6	22	20	$+2$	$4/20 = 0.20$
Total	**120**	**120**	**0**	$\chi^2_{\text{cal}} = \mathbf{4.70}$

Decision: With $\nu = 6 - 1 = 5$ d.f., $\chi^2_{\text{crit}} = 11.07$. Since $\chi^2_{\text{cal}} = 4.70 < 11.07$, we fail to reject H_0. The die is consistent with being unbiased.

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7. Chi-Square Test for Independence of Attributes

7.1 $r \times c$ Contingency Tables

Contingency Table and Expected Cell Frequency

When N observations are classified simultaneously across two qualitative attributes A (with r levels $A_1, \dots, A_r$) and B (with c levels $B_1, \dots, B_c$), the frequencies form an $r \times c$ contingency table.
Null Hypothesis: H_0: Attributes A and B are independent.
Under H_0, $P(A_i \cap B_j) = P(A_i)P(B_j)$, yielding the **expected cell frequency** E_{ij}:

$$E_{ij} = N \times \left(\frac{R_i}{N}\right) \times \left(\frac{C_j}{N}\right) = \frac{R_i \times C_j}{N} \tag{32}$$

where R_i is the i-th row total, C_j is the j-th column total, and N is the grand total. The test statistic is:

$$\chi^2 = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \tag{33}$$

with degrees of freedom:

$$\nu = (r-1)(c-1) \tag{34}$$

7.2 2×2 Contingency Table and Yates' Continuity Correction

2×2 Contingency Table and Yates' Continuity Correction

For a 2×2 contingency table with cell counts:

Attribute	B_1	B_2	Row Total
A_1	a	b	$a+b$
A_2	c	d	$c+d$
Column Total	$a+c$	$b+d$	$N = a+b+c+d$

With $\nu = (2-1)(2-1) = 1$, the standard and **Yates' continuity-corrected** (when $E_{ij} < 5$) statistics are:

$$\chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}, \qquad \chi^2_{\text{corr}} = \frac{N\left(|ad-bc| - \frac{N}{2}\right)^2}{(a+b)(c+d)(a+c)(b+d)} \tag{35}$$

Yates' correction subtracts $N/2$ to account for discrete counts approximating continuous χ^2, preventing inflated Type I error.

Worked Example: 2×2 Contingency Table with Yates' Correction

In a medical study to evaluate the effectiveness of a new vaccine, 200 subjects were observed:

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Group	Infected	Not Infected	Total
Vaccinated	$a = 12$	$b = 88$	$a + b = 100$
Unvaccinated	$c = 28$	$d = 72$	$c + d = 100$
Total	$a + c = 40$	$b+d = 160$	$N = 200$

Test at the 5% level of significance whether the vaccine is significantly effective ($\chi^2_{0.05,1} = 3.841$).

Solution:

1. **Hypotheses:** H_0: Vaccination and infection are independent (no effect) vs. H_1: The vaccine is effective.
2. **Cross-Product Difference:** $|ad - bc| = |(12)(72) - (88)(28)| = |864 - 2464| = 1600$.
3. **Yates' Corrected Test Statistic ($\nu = 1$):**

$$\chi^2_{\text{corr}} = \frac{200 \times (1600 - 100)^2}{100 \times 100 \times 40 \times 160} = \frac{200 \times (1500)^2}{64,000,000} = \frac{450,000,000}{64,000,000} \approx 7.031$$

4. **Conclusion:** Since $\chi^2_{\text{corr}} = 7.031 > \chi^2_{0.05,1} = 3.841$, we reject H_0 at the 5% level. The vaccine significantly reduces infection rates.

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Quick Revision — Module 6

Small Samples: t-Test, F-Test, and Chi-Square Distribution | BS-M301

Key Small Sample and Chi-Square Formulas at a Glance

Single Mean t-Test	$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}, \quad s^2 = \frac{1}{n-1}\sum(x_i-\bar{x})^2, \quad \nu = n-1$
Two-Sample Mean t-Test	$t = \frac{\bar{x}_1-\bar{x}_2}{s_p\sqrt{1/n_1+1/n_2}}, \quad s_p^2 = \frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}, \quad \nu = n_1+n_2-2$
Paired t-Test	$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \quad d_i = x_i - y_i, \quad s_d^2 = \frac{1}{n-1}\sum(d_i-\bar{d})^2, \quad \nu = n-1$
Correlation t-Test	$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}, \quad \nu = n-2$
Fisher's F-Statistic	$F = \frac{s_1^2}{s_2^2} \geq 1, \quad \nu_1 = n_1 - 1, \quad \nu_2 = n_2 - 1$
χ^2 Goodness of Fit	$\chi^2 = \sum_{i=1}^{k} \frac{(O_i-E_i)^2}{E_i}, \quad \nu = k-1-m$
2×2 Contingency χ^2	$\chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}, \quad \nu = 1$
Yates' Correction	$\chi^2_{\text{corr}} = \frac{N(\lvert ad-bc\rvert - N/2)^2}{(a+b)(c+d)(a+c)(b+d)}$

One-Line Definitions

- **Small Sample:** A statistical sample with size $n < 30$, requiring exact distributions (t, F, χ^2) rather than the standard normal Z-test.
- **Degrees of Freedom (ν):** The number of unconstrained, freely variable observations used in evaluating a sample statistic ($\nu = n - k$).
- **Student's t-Distribution:** The exact distribution of the ratio of a standard normal variable to the square root of an independent χ^2 variable divided by its d.f.
- **Pooled Variance (s_p^2):** The weighted average of two independent sample variances used in two-sample t-tests under equal population variances.
- **Paired t-Test:** A test applied to dependent matched pairs where differences $d_i = x_i - y_i$ eliminate extraneous subject variability.
- **Snedecor's F-Distribution:** The distribution of the ratio of two independent chi-square variables, each divided by its d.f. (s_1^2/s_2^2).
- **Chi-Square (χ^2) Distribution:** The distribution formed by the sum of squares of ν independent standard normal random variables.
- **Goodness of Fit:** A statistical test measuring whether an empirical sample frequency distribution adheres to a theoretical distribution.
- **Contingency Table:** A grid matrix displaying the bivariate joint frequency distribution of two qualitative attributes.
- **Yates' Correction:** A continuity correction subtracting $N/2$ from $|ad - bc|$ in 2×2 tables to account for discrete counts in continuous χ^2.

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Frequently Asked / Viva Questions (Part 1)

1. **Q1. Why can we not use the standard normal Z-test for small samples ($n < 30$)?**

   For $n < 30$, the Central Limit Theorem cannot guarantee normality of the sampling distribution unless the parent population is normal. Furthermore, substituting the sample variance s^2 for the unknown σ^2 introduces substantial additional sampling variability, making the tails of the distribution much heavier than $N(0,1)$. Gosset's t-distribution accounts precisely for this extra variability.

2. **Q2. What are the key assumptions underlying the two-sample independent t-test?**

   The assumptions are: (1) both parent populations are normally distributed, (2) the two samples are selected randomly and independently, and (3) the two population variances are equal ($\sigma_1^2 = \sigma_2^2 = \sigma^2$, known as homoscedasticity). Equality of variances is checked using an F-test.

3. **Q3. When should a paired t-test be used instead of an independent two-sample t-test?**

   A paired t-test must be used when the two sets of observations are intrinsically dependent, such as measurements on the same physical subjects before and after a treatment, or twin studies. It isolates the treatment effect by converting paired data into single differences $d_i = x_i - y_i$, thereby eliminating between-subject biological or physical variance.

4. **Q4. State the relation between Student's t, Snedecor's F, and Chi-square χ^2 distributions.**

   Key interrelations include:

   - The square of a t-variable with ν d.f. is an F-variable with $(1, \nu)$ d.f.: $t_\nu^2 = F_{1,\nu}$.
   - As the denominator degrees of freedom $\nu_2 \to \infty$, $\nu_1 F_{\nu_1,\nu_2} \to \chi^2(\nu_1)$.
   - As degrees of freedom $\nu \to \infty$, Student's $t(\nu) \to N(0,1)$.

5. **Q5. Why is the larger sample variance always placed in the numerator in an F-test?**

   By convention, setting $F = s_1^2/s_2^2$ with $s_1^2 \geq s_2^2$ ensures $F \geq 1$. This allows the use of standard upper-tail F-distribution critical value tables without needing to invert lower-tail critical values ($F_{1-\alpha} = 1/F_\alpha$).

6. **Q6. What are the fundamental validity conditions for Karl Pearson's Chi-square test?**

   The conditions are: (1) total sample size $N = \sum O_i \geq 50$, (2) each theoretical expected frequency $E_i \geq 5$ (if $E_i < 5$, adjacent cells must be pooled), (3) individual observations are drawn randomly and independently, and (4) $\sum O_i = \sum E_i = N$.

7. **Q7. How are degrees of freedom calculated in an $r \times c$ contingency table?**

   The degrees of freedom are $\nu = (r-1)(c-1)$. This arises because the marginal row totals impose r constraints and the column totals impose c constraints, with one common grand total constraint N, leaving $(r-1)(c-1)$ independently assignable cells.

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Frequently Asked / Viva Questions (Part 2)

8. **Q8. What is Yates' correction for continuity and when is it mandatory?**

   Yates' correction adjusts the discrete cell counts in a 2×2 table to match the continuous theoretical χ^2 density by subtracting $N/2$ from the absolute difference $|ad - bc|$. It is mandatory when $\nu = 1$ and any cell frequency is less than 5 ($E_{ij} < 5$), preventing spurious rejections of H_0.

9. **Q9. How do you test whether an observed sample correlation coefficient $r = 0.6$ from $n = 18$ pairs is statistically significant?**

   We test $H_0 : \rho = 0$ using $t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.6\sqrt{16}}{\sqrt{1-0.36}} = \frac{0.6\times 4}{0.8} = 3.00$. With $\nu = 18 - 2 = 16$ d.f., the critical value at $\alpha = 0.05$ is $t_{0.025,16} = 2.12$. Since $|t_{\text{cal}}| = 3.00 > 2.12$, the correlation is statistically significant.

10. **Q10. What does a non-significant result in a χ^2 goodness-of-fit test signify?**

    When $\chi^2_{\text{cal}} \leq \chi^2_{\text{crit}}$, we fail to reject H_0. This indicates that the observed sampling discrepancies from theoretical frequencies are small enough to be attributed purely to chance, confirming the theoretical model fits the empirical data well.

Summary Matrix: Comparison of Small Sample Tests (t, F, χ^2)

Test	Objective	Test Statistic & Parameters	d.f. (ν)
Single Mean t	**Test** $H_0 : \mu = \mu_0$ **(σ unknown)**	$t = \dfrac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad s^2 = \dfrac{\sum(x_i - \bar{x})^2}{n-1}$	$n - 1$
Two-Sample t	**Compare two means ($\sigma_1 = \sigma_2$)**	$t = \dfrac{\bar{x}_1 - \bar{x}_2}{s_p\sqrt{1/n_1 + 1/n_2}}$, pooled s_p^2	$n_1 + n_2 - 2$
Paired t	**Before vs. After on same units**	$t = \dfrac{\bar{d}}{s_d/\sqrt{n}}$, differences $d_i = x_i - y_i$	$n - 1$
Correlation t	**Test** $H_0 : \rho = 0$	$t = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}}$	$n - 2$
Fisher's F	**Compare σ_1^2 vs σ_2^2**	$F = \dfrac{s_1^2}{s_2^2}$ $(s_1^2 \geq s_2^2)$	$(n_1 - 1, n_2 - 1)$
Pearson's χ^2	**Goodness of Fit**	$\chi^2 = \sum \dfrac{(O_i - E_i)^2}{E_i}$	$k - 1 - m$
Contingency χ^2	**Independence of attributes**	$\chi^2 = \sum\sum \dfrac{(O_{ij} - E_{ij})^2}{E_{ij}}$ or 2×2 Yates	$(r-1)(c-1)$

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